

ECE 361 HW 2

Colin Gregg

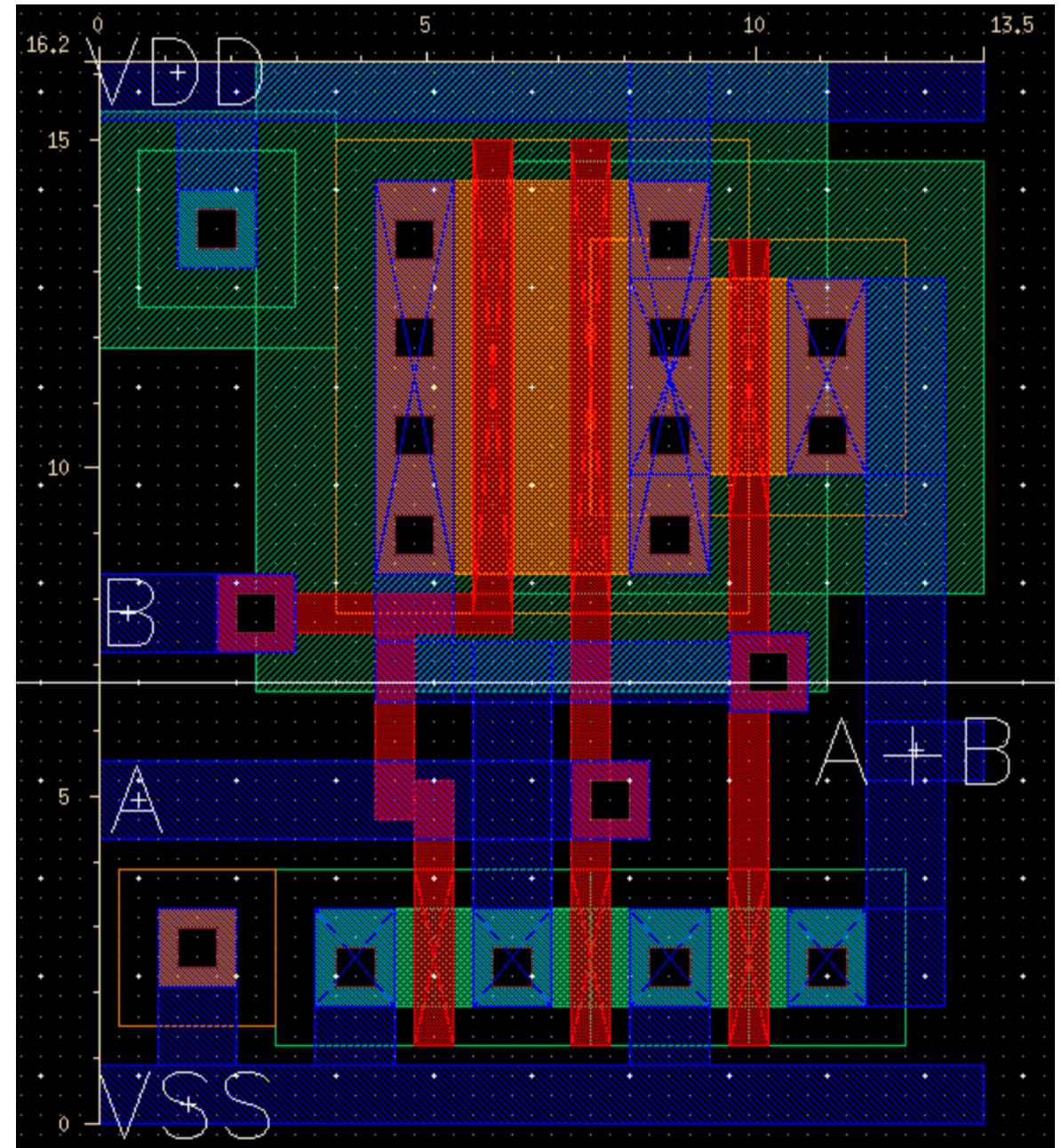
Guidelines to Follow:

- Create OR-gate using 1 NOR-gate and 1 Inverter-gate
- PMOS in NOR-gate has width of 6um
- PMOS in Inverter-gate has width of 3um
- All NMOS have width of 1.5um
- **Goal is to minimize area of device and parasitic cap./res.**

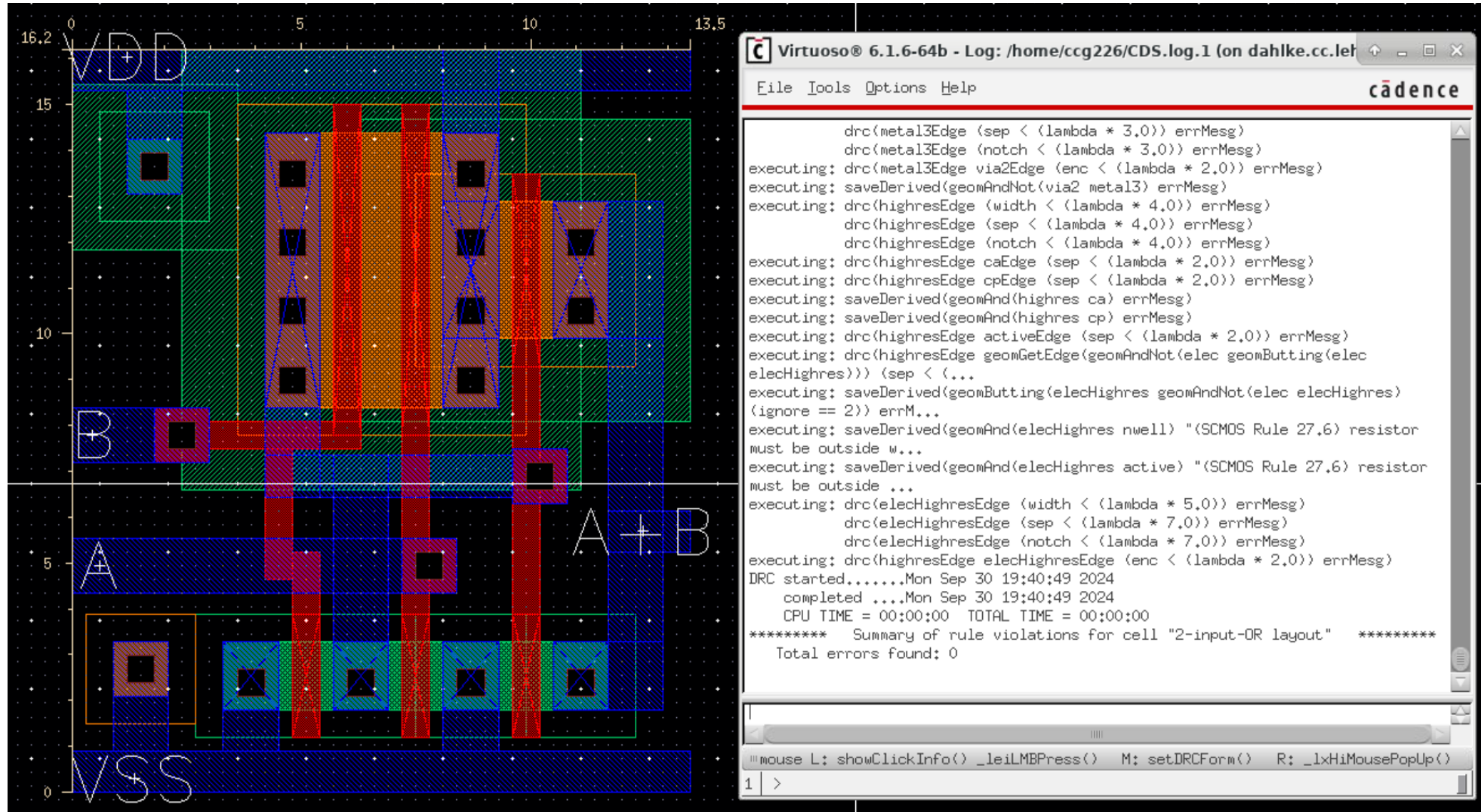
Design:

$$\text{Area} = (16.2 \times 13.5)(\mu\text{m})^2$$

$$\text{Area} = 218.7 (\mu\text{m})^2$$



DRC Check



The image displays a Virtuoso CAD interface with a DRC (Design Rule Check) window open on the right. The main window shows a circuit board layout with various colored regions (blue, green, orange, red) and labels like 'VDD', 'VSS', 'A', 'B', and 'A+B'. The DRC window shows the results of the check, including a list of rules and their violations.

Virtuoso® 6.1.6-64b - Log: /home/ccg226/CDS.log.1 (on dahlke.cc.leh)

File Tools Options Help

cadence

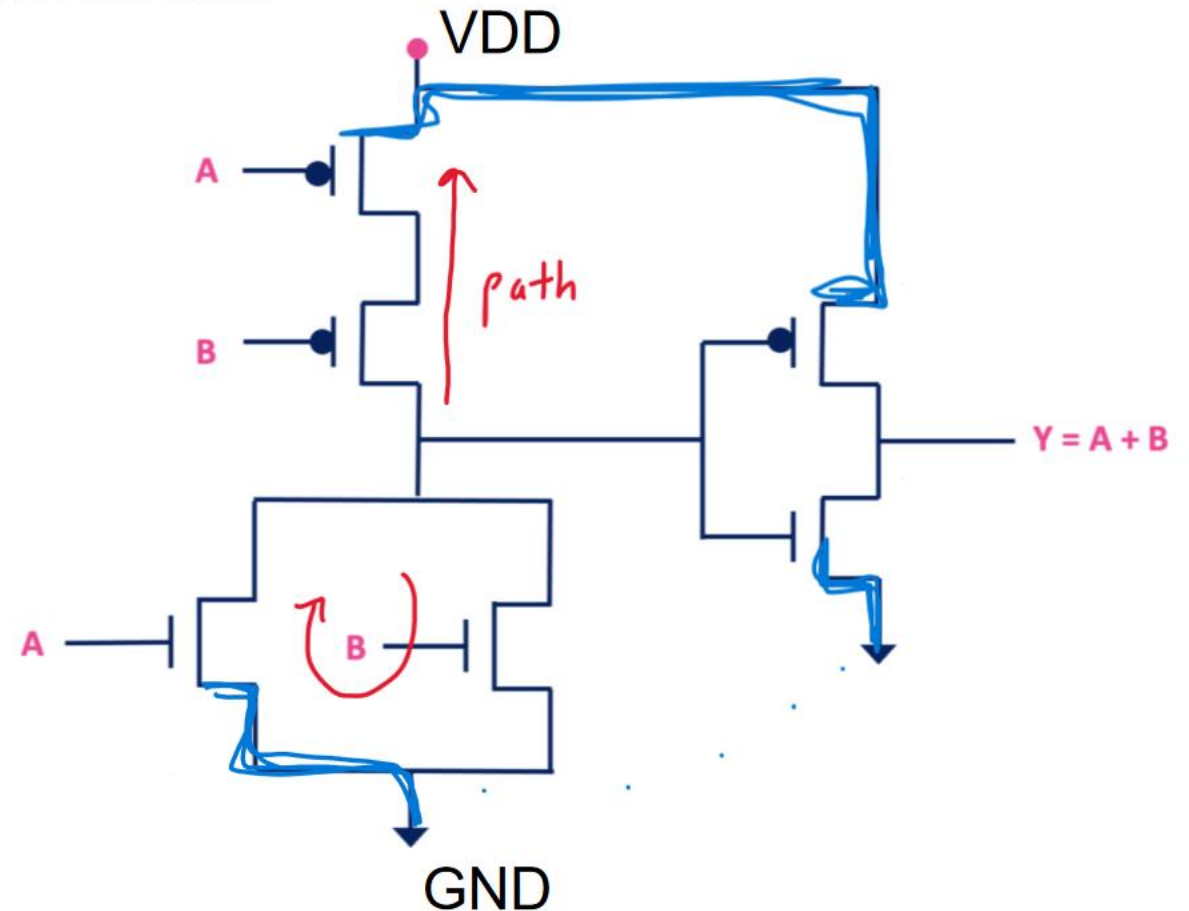
```
drc(metal3Edge (sep < (lambda * 3.0)) errMesg)
drc(metal3Edge (notch < (lambda * 3.0)) errMesg)
executing: drc(metal3Edge via2Edge (enc < (lambda * 2.0)) errMesg)
executing: saveDerived(geomAndNot(via2 metal3) errMesg)
executing: drc(highresEdge (width < (lambda * 4.0)) errMesg)
drc(highresEdge (sep < (lambda * 4.0)) errMesg)
drc(highresEdge (notch < (lambda * 4.0)) errMesg)
executing: drc(highresEdge caEdge (sep < (lambda * 2.0)) errMesg)
executing: drc(highresEdge cpEdge (sep < (lambda * 2.0)) errMesg)
executing: saveDerived(geomAnd(highres ca) errMesg)
executing: saveDerived(geomAnd(highres cp) errMesg)
executing: drc(highresEdge activeEdge (sep < (lambda * 2.0)) errMesg)
executing: drc(highresEdge geomGetEdge(geomAndNot(elec geomButting(elec
elecHighres))) (sep < (...
executing: saveDerived(geomButting(elecHighres geomAndNot(elec elecHighres)
(ignore == 2)) errM...
executing: saveDerived(geomAnd(elecHighres nwell) "(SCMOS Rule 27,6) resistor
must be outside w...
executing: saveDerived(geomAnd(elecHighres active) "(SCMOS Rule 27,6) resistor
must be outside ...
executing: drc(elecHighresEdge (width < (lambda * 5.0)) errMesg)
drc(elecHighresEdge (sep < (lambda * 7.0)) errMesg)
drc(elecHighresEdge (notch < (lambda * 7.0)) errMesg)
executing: drc(highresEdge elecHighresEdge (enc < (lambda * 2.0)) errMesg)
DRC started.....Mon Sep 30 19:40:49 2024
completed ....Mon Sep 30 19:40:49 2024
CPU TIME = 00:00:00 TOTAL TIME = 00:00:00
***** Summary of rule violations for cell "2-input-OR layout" *****
Total errors found: 0
```

mouse L: showClickInfo() _leiLMBPress() M: setDRCForm() R: _lxHiMousePopUp()

1 >

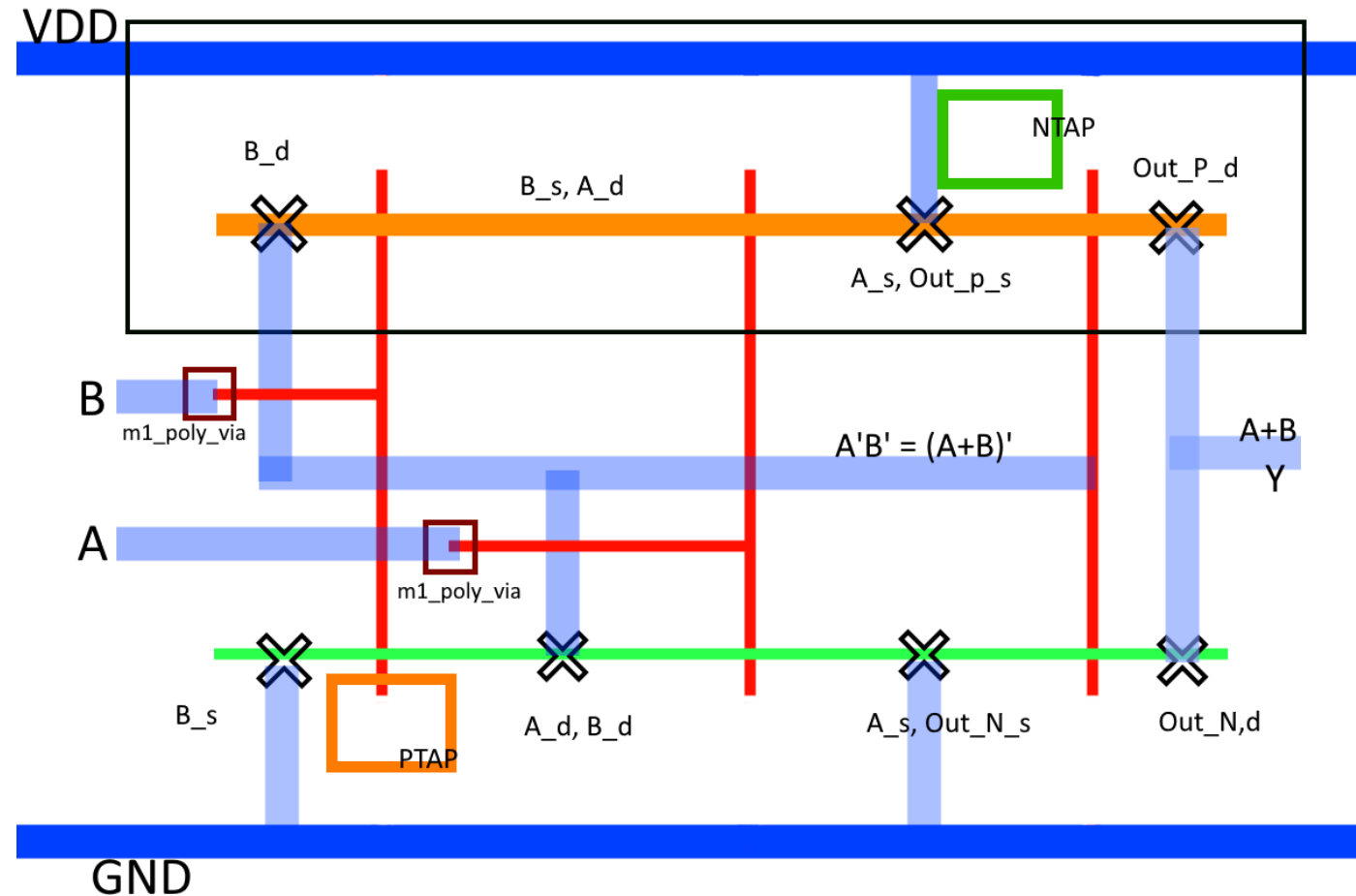
Schematic

- Following principle in textbook, created path in PUN, and replicated path in PDN, for NOR circuit.
- Noticed A source for PMOS and NMOS in NOR connected to equivalent in Inverter
 - For this reason, made path start with B so that A input would be closer to inverter and could utilize common sources



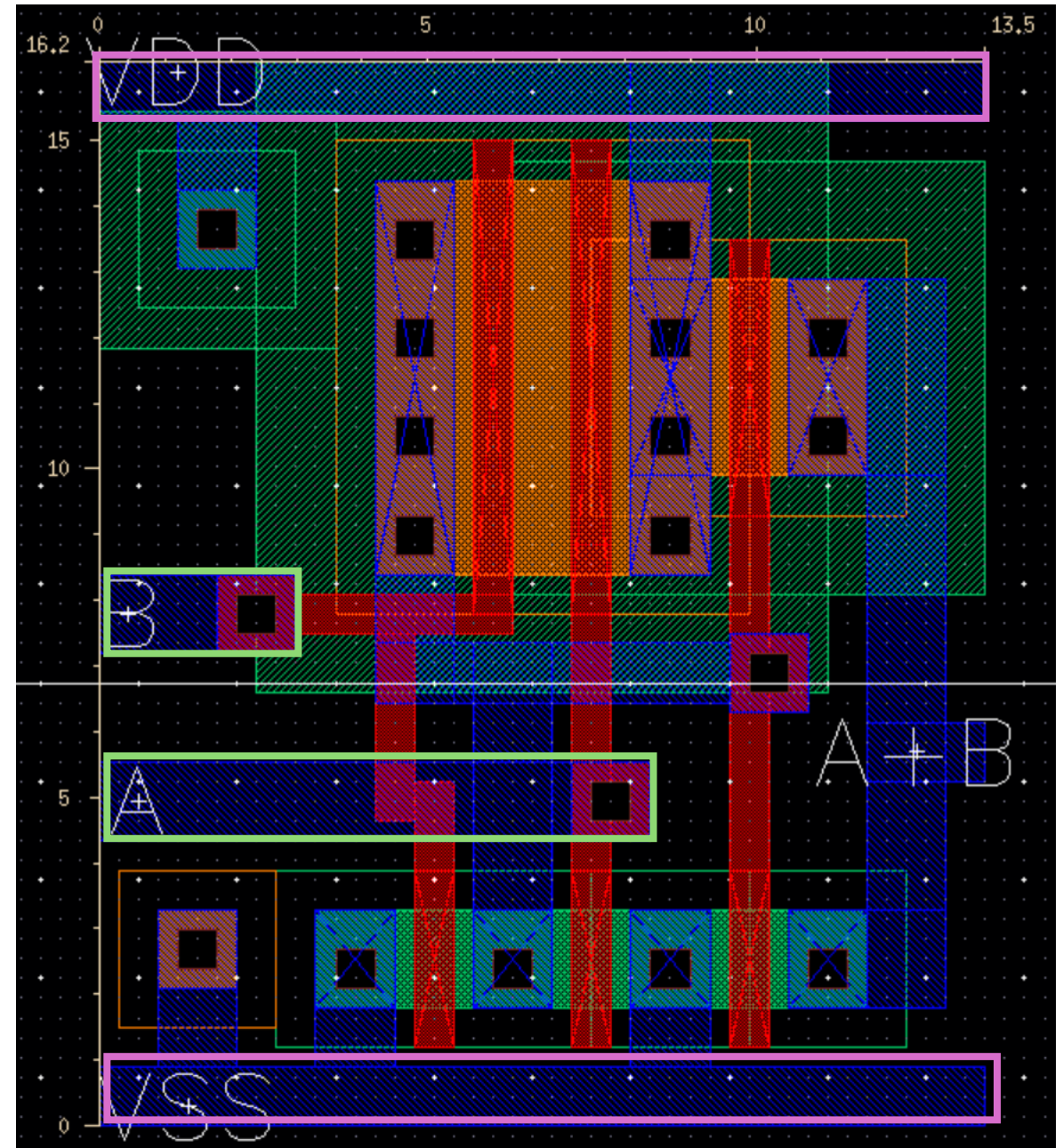
Stick Drawing

- From schematic, created stick drawing
- Note: A input's common sources with inverter's transistors



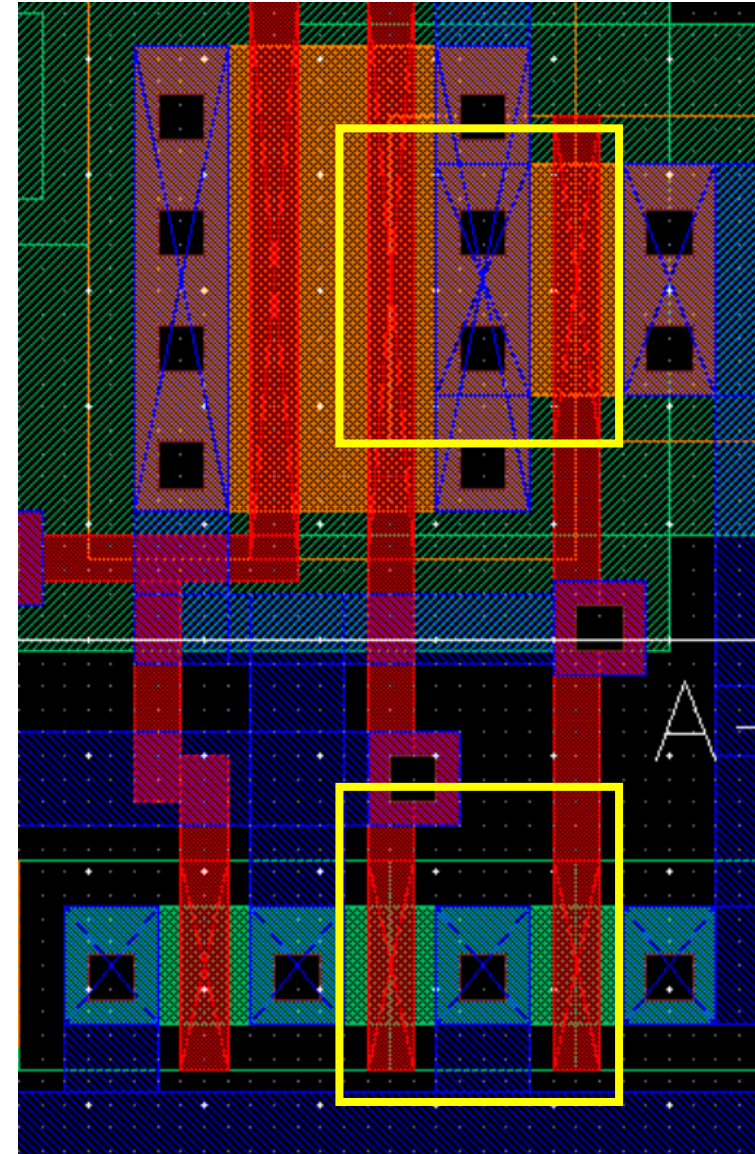
Layout (con't)

- The inputs A and B have been highlighted (in green)
- VDD and VSS have also been highlighted (in magenta)



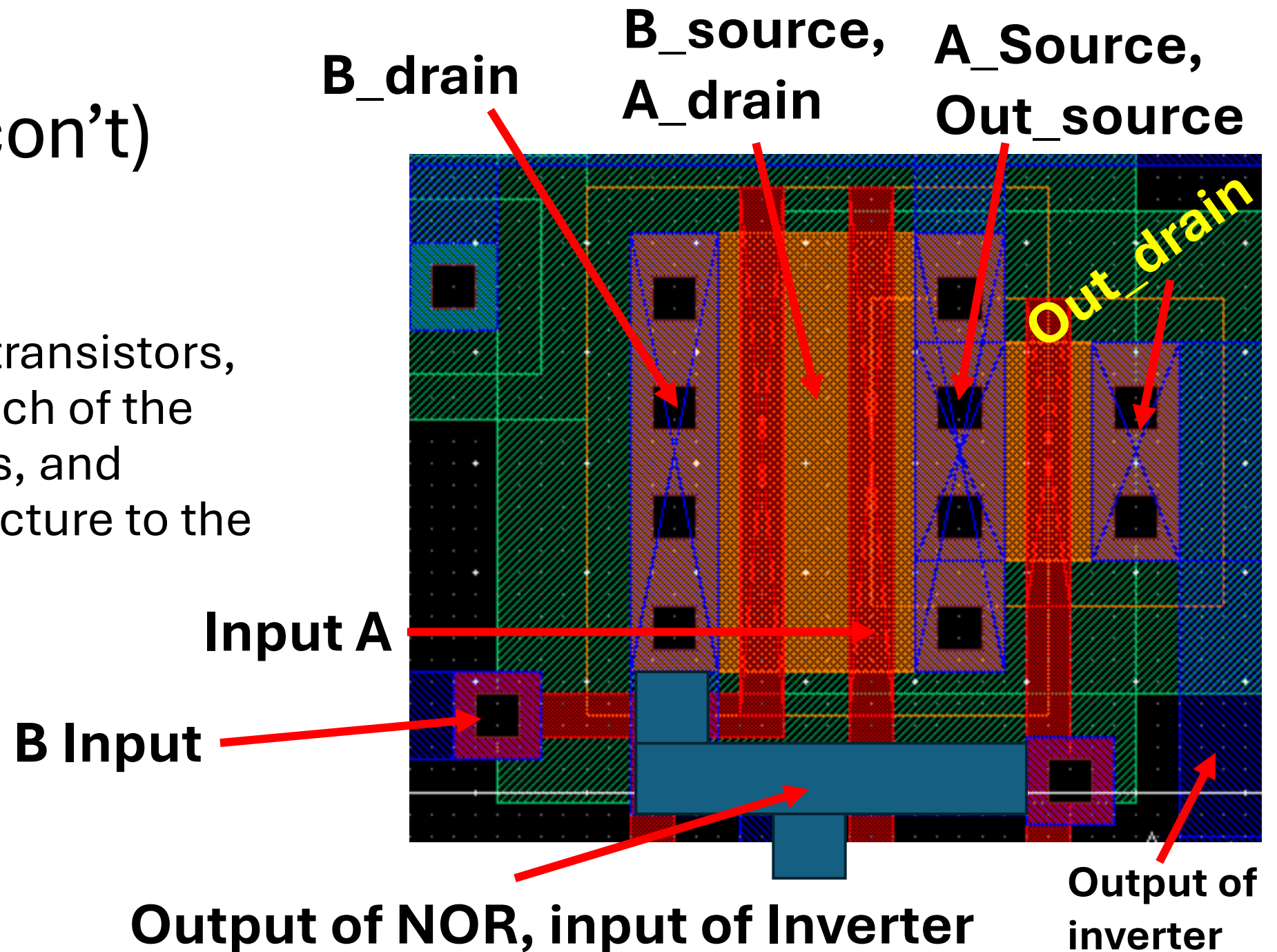
Layout (con't)

- In yellow rectangle we note the common sources between the A input in the NOR gate and the transistors in the Inverter
- To achieve this in the layout
 - for the PMOS: dragged the two different sized PMOS's over each other so that the contacts for m1 to active overlapped flush
 - For NMOS: I only used 1 transistor with a multiplier of 3



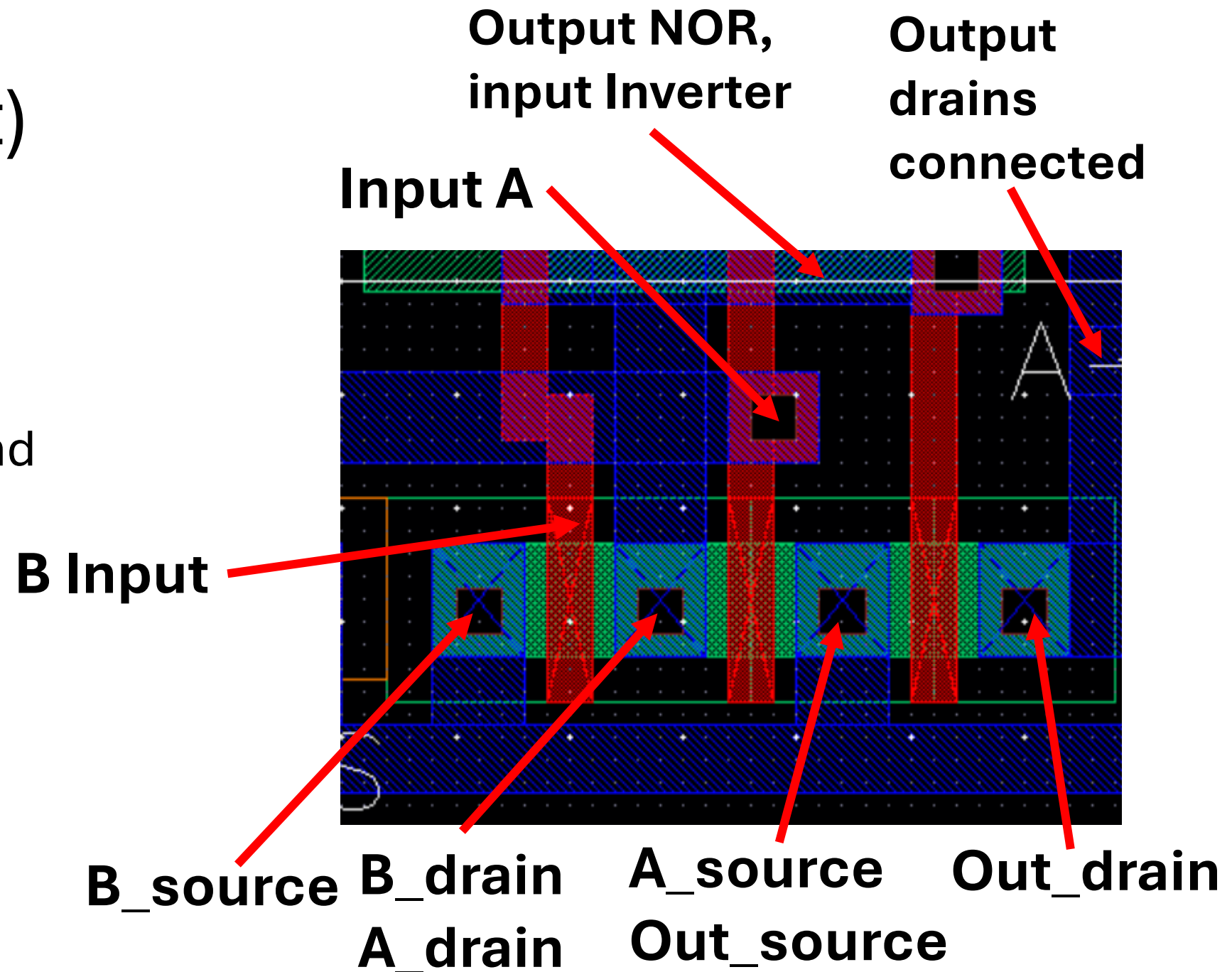
Layout (con't)

For the PMOS transistors, you can see each of the drains, sources, and inputs in the picture to the right:



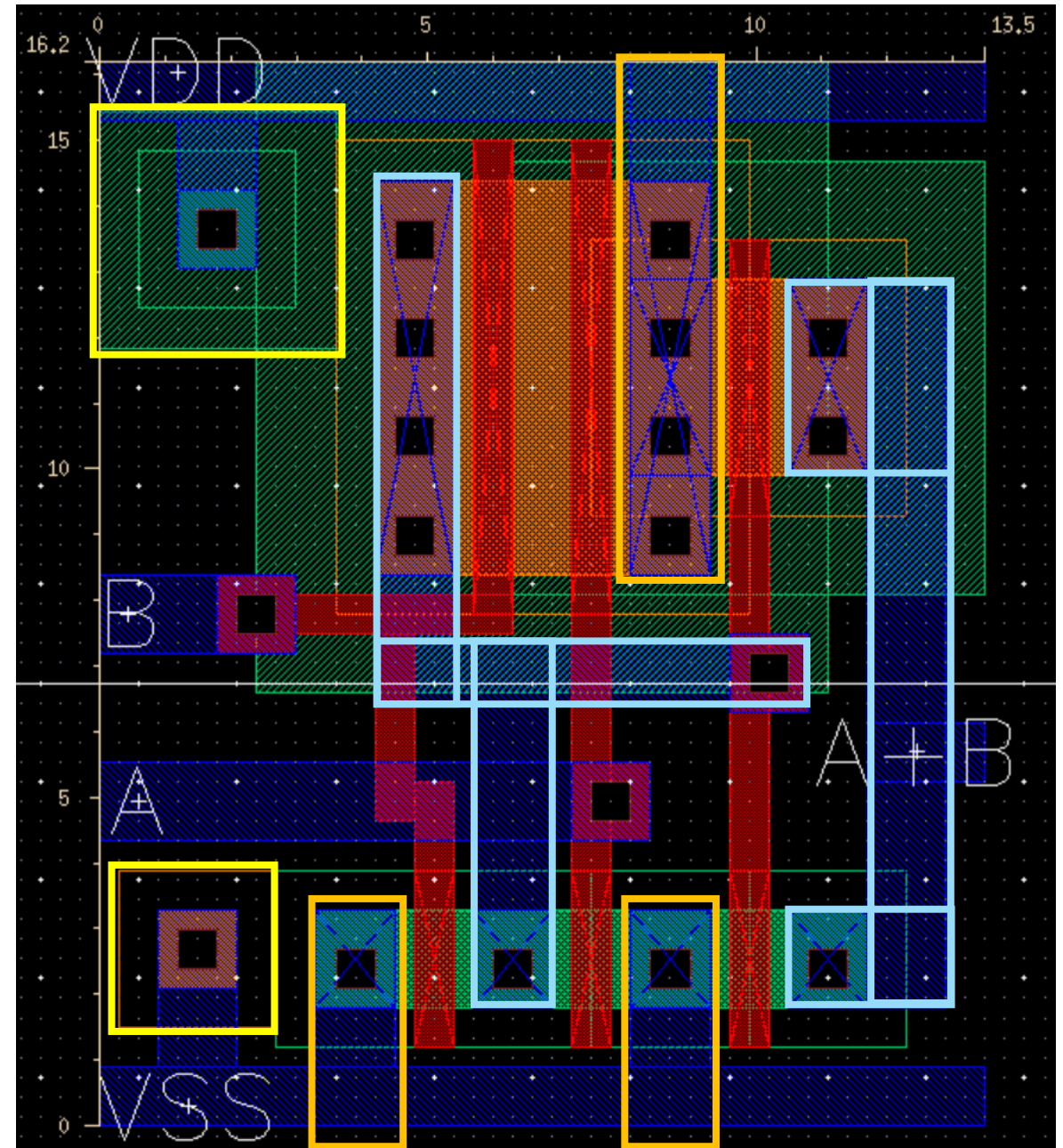
Layout (con't)

- The NMOS drains, sources, inputs, and outputs:



Layout (con't)

- I have highlighted in yellow the PTAP (connected VSS) and NTAP (connected VDD)
- I have also highlighted the NMOS source connections to VSS and PMOS source connection to VDD (in orange)
- Lastly, I highlighted the NOR common sources and Inverter common drains carrying each circuit's output (in blue)



Thank You